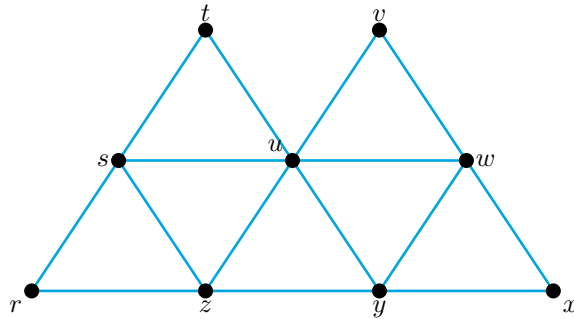
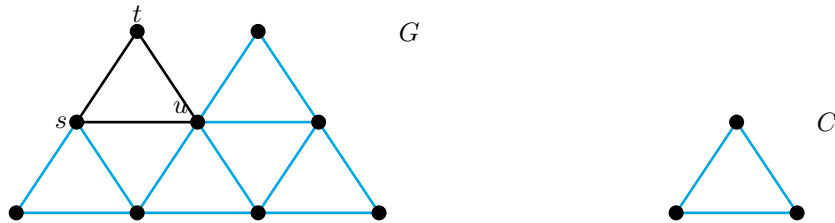


10.1.15

Determine which of the graphs in 12-17 have Euler circuits. If the graph does not have an Euler circuit, explain why not. If it does have an Euler circuit, describe one.

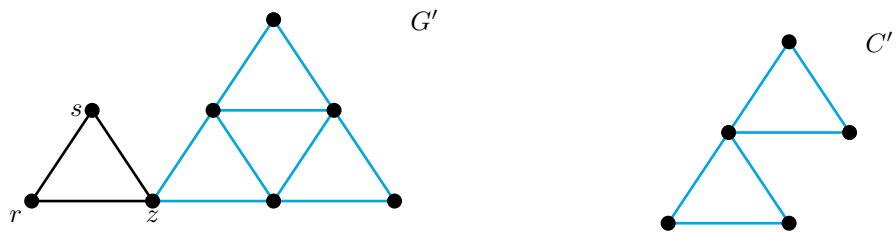


If the graph G is connected and the degree of every vertex of G is a positive even integer, then the graph has an Euler circuit. By this theorem, we can pick any vertex v at which to start, and construct a circuit C , which is a sequence of adjacent vertices and edges, starting and ending at v , and never repeating an edge.



$$\deg_G(s) = 4$$

If C does not contain every edge and vertex of G , we can remove all edges of C from G , and also any vertices that become isolated when the edges of C are removed. This becomes a subgraph G' . We then pick any vertex common to both C and G' , and continue picking any sequence of adjacent vertices and edges.



$$\deg_G(s) = 4 \rightarrow \deg_{G'}(s) = 2$$

Since a circuit enters and leaves each vertex, removing a circuit removes two incident edges each time it passes through a vertex. Therefore, removing C preserves the even degree of the remaining vertices. This allows us to construct another circuit in G' .

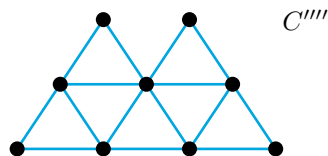
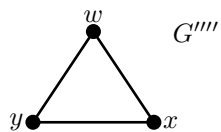


$$\deg_{G'}(s) = 2 \rightarrow \deg_{G''}(s) = 0$$

Now, s is no longer a vertex of G'' , but we may continue constructing circuits since every remaining vertex has positive even degree.



$$\deg_{G''}(u) = 4 \rightarrow \deg_{G'''}(u) = 2$$



$$\deg_{G'''}(u) = 2 \rightarrow \deg_{G'''}(u) = 0$$

The final subgraph concludes the construction, since every vertex has had positive even degree throughout the process. Every sequence of adjacent vertices and edges has been incorporated into the circuit without repeating an edge, until every edge of the original graph has been used. Therefore, the graph has an Euler circuit.

Solution: The graph has an Euler circuit.